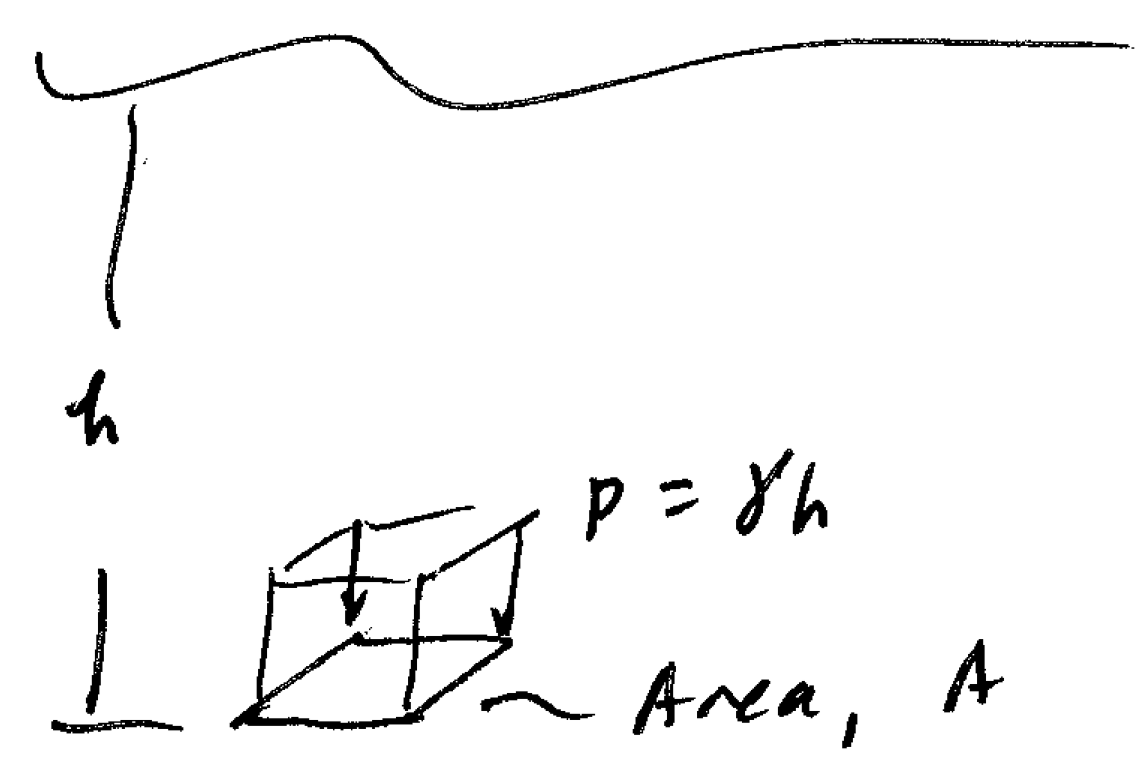
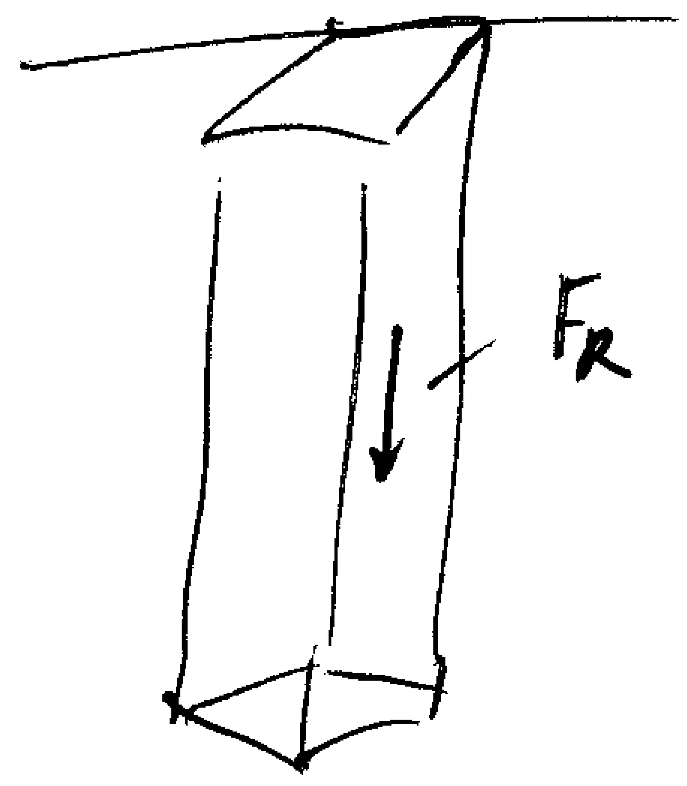
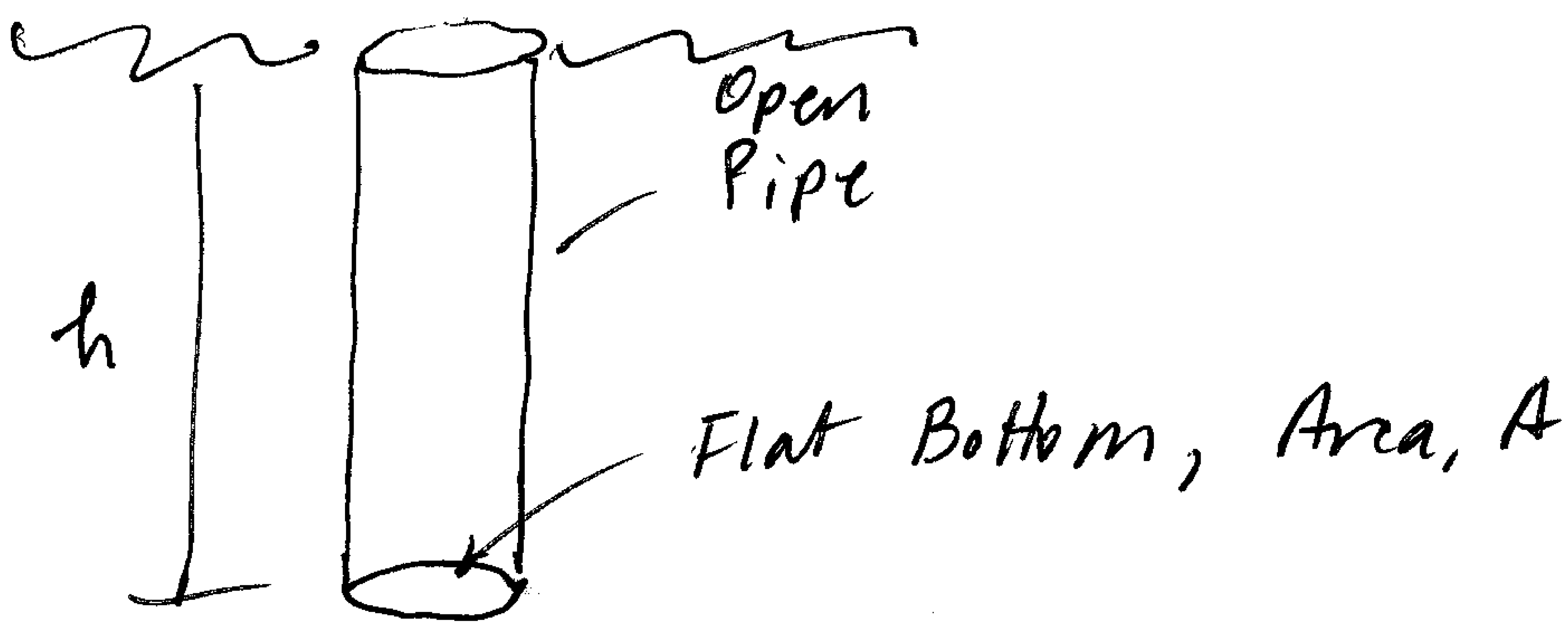
Force on Horiz Plate



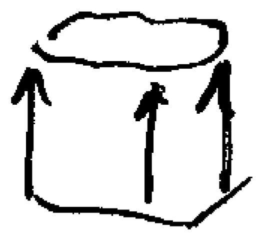
$F_R = \gamma \cdot h\,(A)$



But,
$h \cdot A$ = Volume of Column
and $\gamma \cdot h \cdot A = \gamma \cdot$ Volume
= Weight of column of fluid !



↑ F_R upward on flat bottom of pipe ?

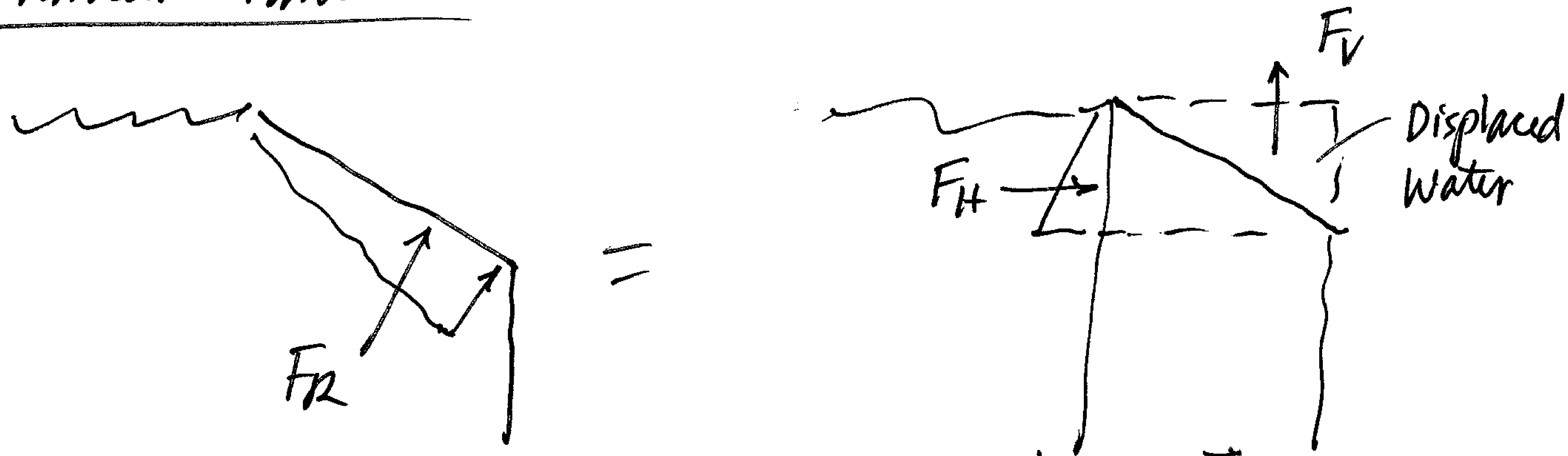


Pressure $p = \gamma \cdot h$
acts on Area, A

$+\uparrow \quad F_R = p \cdot \text{Area} = \gamma \cdot h \cdot A = \gamma \cdot \text{Volume}$

But again, $h \cdot A$ = Volume of pipe !
= Volume (Wt) of Fluid displaced by the pipe.

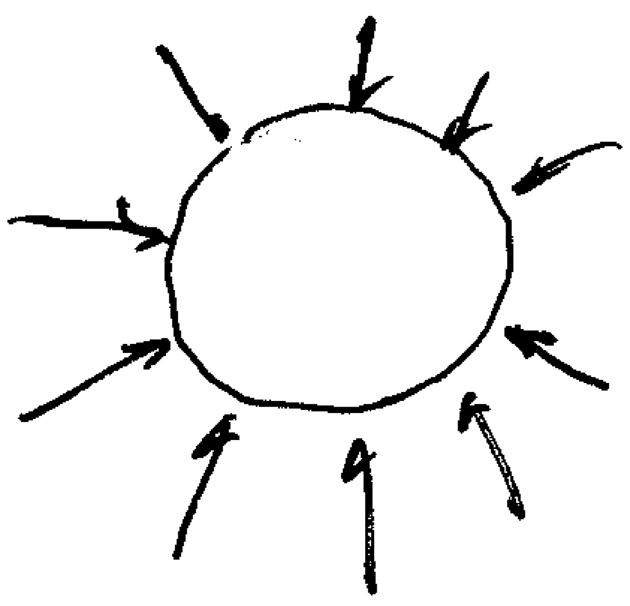
Slanted Tank



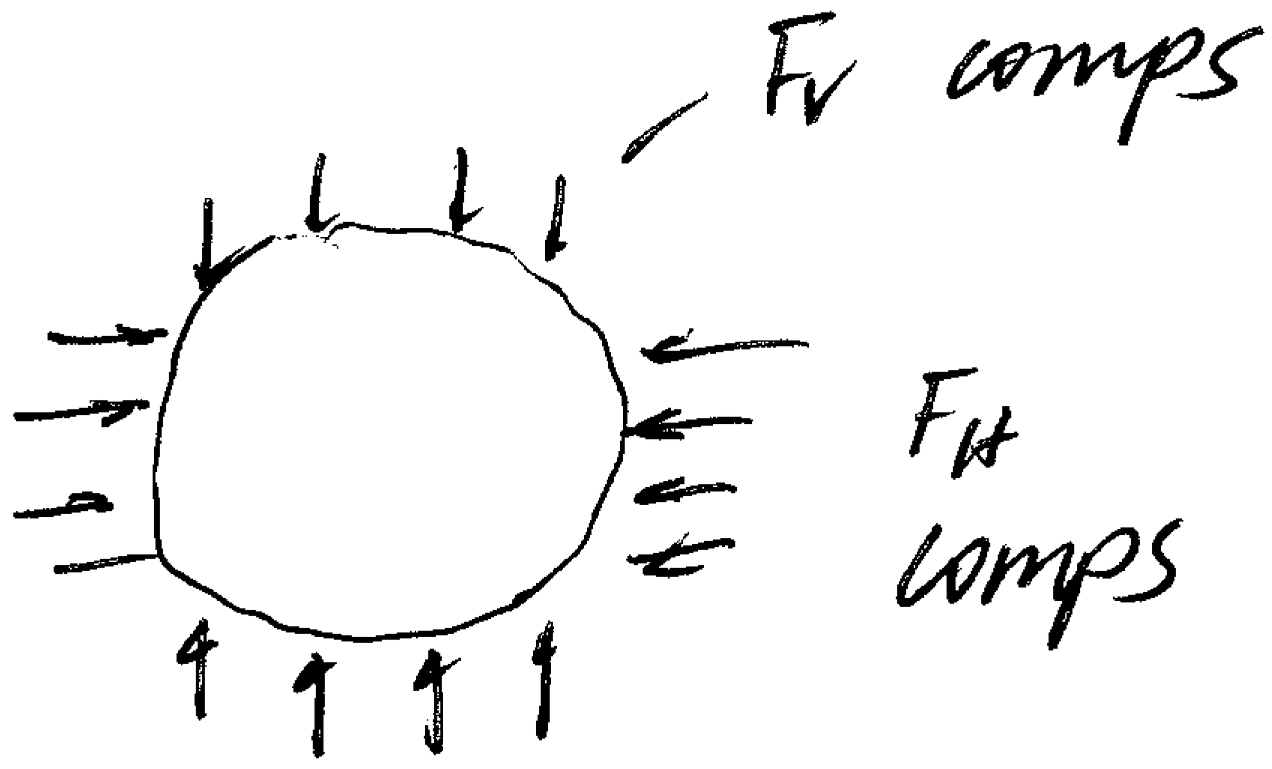
$$\vec{F_R} = \vec{F_H} + \vec{F_V}$$

Archimedes

Bouyancy Force = Wt of fluid displaced by a submerged object

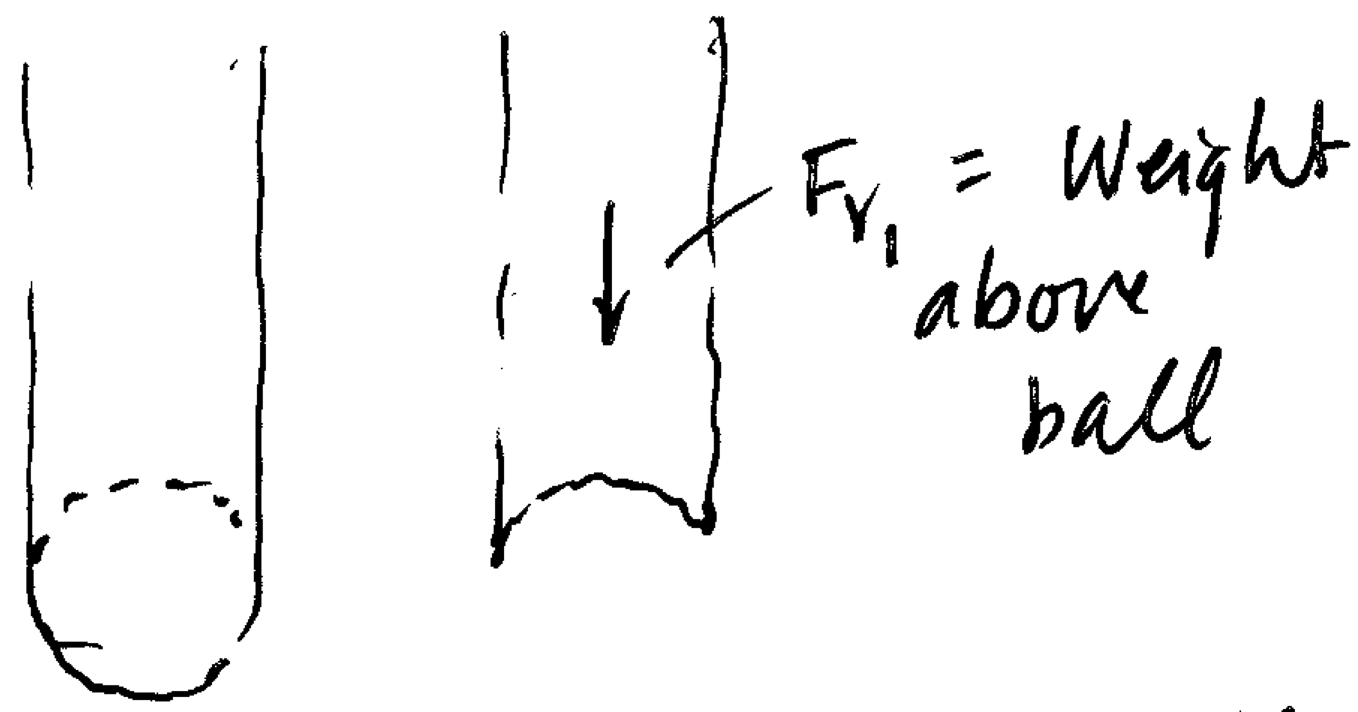


Pressure acts $\perp$



Resolve into F_H and F_V comp's

F_H comps cancel one another.



$\uparrow F_{V_2}$ = Total Volume $\cdot \gamma$

$F_{V_2} > F_{V_1}$ $\quad$ $F_{V_2} - F_{V_1}$ = Wt of fluid displaced by object